

Matthew Butrovich

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Summary

Database management systems (DBMS) expert with a deep understanding of high-performance distributed data processing. Strong background in transactional and analytical DBMS design (query execution, storage management, concurrency control, data formats), data lakes (Iceberg, Parquet), Linux internals (eBPF, networking), storage systems (distributed, devices, file systems), performance engineering (benchmarking, observability), systems for machine learning, and machine learning for systems. Committed to lifelong learning, peer mentorship, and contributing to scientific and open source communities.

Professional Experience

2024 –	Apple Senior Distributed Database Engineer	Washington, DC
	- Enhancing Apple’s data lakehouse by contributing to multiple open source Apache projects: Arrow, DataFusion, DataFusion Comet, Iceberg, Parquet, and Spark. - Brought Comet to production-quality, reducing the cost of Spark pipelines by >50%. - Enabled Comet’s dynamic partition pruning (adaptive and non-adaptive query execution), which previously forced queries to fall back to slower Spark execution. Across 78 TPC-DS benchmark queries, the share of each query running natively rose from ~30-50% to ~95-97%, contributing to a >50% overall speedup at 1 TB. - Designed and shipped Comet’s native Iceberg scan, upstreaming >30 PRs to Iceberg. - Drove a DataFusion sort-merge-join (SMJ) overhaul across 12 PRs, making TPC-H >500x faster and near-unique LEFT/FULL joins >20x faster. - Diagnosed and fixed a Comet busy-poll regression, dropping cluster CPU utilization from 30% to 8% on I/O-bound Iceberg workloads. - Re-architected the JVM-to-native data path to true zero-copy using the Arrow C Stream Interface, eliminating a per-batch deep copy on every batch crossing the boundary.	
2009 – 2017	Western Digital Corporation Lead Technician V	Irvine, CA
	- Led a team of 17 technicians verifying firmware for HDD and SSD storage devices. - Promoted four times from initial role as Technician II while achieving B.Sc. degree.	

Education

2019 – 2024	Carnegie Mellon University Doctor of Philosophy (Ph.D.), Computer Science	Pittsburgh, PA
	- Advisor: Andy Pavlo - Thesis: “On Embedding Database Management System Logic in Operating Systems via Restricted Programming Environments”	
2017 – 2018	Carnegie Mellon University Master of Science (M.Sc.), Computer Science	Pittsburgh, PA
2013 – 2017	University of California, Irvine Bachelor of Science (B.Sc.), <i>cum laude</i> , Computer Science and Engineering	Irvine, CA
2010 – 2013	Irvine Valley College Certificate of Achievement, Computer Science: Computer Languages	Irvine, CA

Selected Talks

2025	CS 224P: Big Data Systems “Fusing Academic Research and Open Source Systems for Petabyte-Scale Analytics”	Link Video
2025	DC Systems “Putting Theory Into Practice”	Link Video
2023	15-445/645: Database Systems Lecture on join algorithms	Link Video
2019	15-445/645: Database Systems Lecture on two-phase locking concurrency control	Link Video

Selected Publications

2025	Matthew Butrovich , Samuel Arch, Wan Shen Lim, William Zhang, Jignesh M. Patel, and Andrew Pavlo. “BPF-DB: A Kernel-Embedded Transactional Database Management System For eBPF Applications”. <i>SIGMOD Conference</i> .
2023	Matthew Butrovich , Karthik Ramanathan, John Rollinson, Wan Shen Lim, William Zhang, Justine Sherry, and Andrew Pavlo. “Tigger: A Database Proxy That Bounces With User-Bypass”. <i>Proc. VLDB Endow.</i>
2022	Matthew Butrovich , Wan Shen Lim, Lin Ma, John Rollinson, William Zhang, Yu Xia, and Andrew Pavlo. “Tastes Great! Less Filling! High Performance and Accurate Training Data Collection for Self-Driving Database Management Systems”. <i>SIGMOD Conference</i> .

Service

To the Open Source Community

2026 –	PMC Member, Apache DataFusion, <i>The Apache Software Foundation</i> .	Link
2025 –	Contributor, Apache Arrow, <i>The Apache Software Foundation</i> .	Link
2025 –	Contributor, Apache Iceberg, <i>The Apache Software Foundation</i> .	Link
2025 –	Contributor, Apache Parquet, <i>The Apache Software Foundation</i> .	Link
2024 –	Contributor, Apache Spark, <i>The Apache Software Foundation</i> .	Link
2025 – 2026	Committer, Apache DataFusion, <i>The Apache Software Foundation</i> .	Link
2024 – 2025	Contributor, Apache DataFusion, <i>The Apache Software Foundation</i> .	Link

To the Profession

2026	Member, <i>ACM SIGMOD 2027 Program Committee</i> .	Link
2026	Member, <i>VLDB 2027 Program Committee</i> .	Link
2026	Member, <i>4th Workshop on eBPF and Kernel Extensions Program Committee</i> .	Link
2025	Member, <i>ACM SIGMOD 2026 Industry Track Program Committee</i> .	Link
2025	Member, <i>ACM SIGMOD 2026 Program Committee</i> .	Link
2025	Reviewer, <i>ACM Transactions on Database Systems (TODS)</i> .	Link
2024	Member, <i>eBPF Summit 2024 Program Committee</i> .	Link
2023	Member, <i>eBPF Summit 2023 Program Committee</i> .	Link
2023	Member, <i>ACM SIGMOD 2023 Artifacts & Reproducibility Committee</i> .	Link

Skills

Programming Languages: C, C++, eBPF, Java, Python, Rust, Scala

Development Tools: bpftrace, Claude, GDB/LLDB, Git, Intel VTune, perf